2 Fundamentals of Discrete and Continuous Time

Fundamentals of signals

Continuous Time

A CT (continuous time) function is periodontic if it repeats itself after a time T

$$X(t) = x(t+T)$$

This is a boundary condition.

$$\begin{aligned} A\cos(\omega_0 t + \phi) &= A\cos(\omega_0(t+T) + \phi) \\ & A\cos(\omega_0 t + \phi + \underbrace{\omega_0 T}_{2\pi}) \end{aligned}$$

This $\omega_0 T$ that gets the function is the fundamental frequency that gets it to repeat.

$$\omega_0 T = 2n\pi,\ \ n = \pm 1, \pm 2 \ldots$$

Where the fundamental period is $T_0 = \frac{2\pi}{\omega_0}$

Lets look at two frequencies added together. What is the fundamental period?

$$x(t) = A\cos(\underbrace{2\pi}_{\omega_0^a} t + \phi_a) + B\cos(\underbrace{3\pi}_{\omega_0^b} t + \phi)$$

$$T_0 = \frac{2\pi n_a}{2\pi} = \frac{2\pi nb}{3\pi}$$

We have to have
$T_0 = n_a = \frac{2}{3} n_b$ at the same time.
This represents how many "cycles" it takes for each function to repeat.
The smallest number for this is $n_a = 2, m_b = 3$.
This means $T_0 = 2$. The fundamental frequency is where the *overall* pattern is repeating.

There are some sums of signals that could *never* be periodic.
For the signals to sync up,
$\frac{n_a}{n_b}$ has to be rational, so $\frac{\omega_0^a}{\omega_0^b}$ would be rational. If one of the ω_0s are irrational, i.e. π, then the sinusoids would NEVER line up perfectly. Weird but cool!

Also, note that a time shift corresponds to a phase shift.

$$A\cos(\omega_0(t+t_0) + \phi) = A\cos(\omega_0 t + \underbrace{(\omega_0 t_0 + \phi)}_{\phi'})$$

$\phi' = \phi + \Delta\phi$, $\Delta\phi = \omega_0 t_0$.

A phase shift could also be expressed as a time shift:

$$\begin{aligned} A\cos(\omega_0 t + \phi) &= a\cos(\omega(t + \Delta t)) \\ \Delta t &= \frac{\phi}{\omega_0} \end{aligned}$$

We're just manipulating the expression. Phase and time shifts are the same.

Now lets move to discrete time.
The frequency

Discrete Time

Lets start with continuous time.
$x(t) = A\cos(\omega_0 t + \phi)$.
We'll sample this at T, the sampling period.

We'll make a discrete signal by

$$x[n] = A\cos(\omega_0 nT + \phi)$$

which is equal to $x(nT)$.
We can define a *dimensionless frequency*, $\hat{\omega}_0 = \omega_0 T$.

$$x[n] = A\cos(\hat{\omega}_0 n + \phi)$$

n is an integer without units of time, so the frequency must also be dimensionless.

Differences

Lets look at other distinctions between CT and DT.

Time and Phase Shifts

A "time" shift in DT is shifting by n, a integer.

$$\begin{aligned} x[n] &= A\cos(\hat{\omega}_0(n+n_0)+\phi) \\ &= A\cos(\hat{\omega}_0 n + \underbrace{\hat{\omega}_0 n_0 + \phi}_{\phi'}) \\ &= A\cos(\hat{\omega}_0 n + \phi') \end{aligned}$$

Can we go the other way?
A time shift is the same as a phase shift.
Are all phase shifts expressible as time shifts?
no!
We can only represent phase shifts that are capturable by integer multiples of $\hat{\omega}_0$.

$$\Delta\phi = \hat{\omega}_0 n_0$$

We can only have phase shifts that are multiples of $\hat{\omega}_0$ because n is an integer.

again: Only phase shifts that are integer multiples of $\hat{\omega}_0$.

Periodicity

$$x[n] = x[n+N]$$

We have a condition on N.

$$\begin{aligned} A\cos(\hat{\omega}_0 n) &= A\cos(\hat{\omega}_0(n+N)) \\ &= A\cos(\hat{\omega}_0 n + \hat{\omega}_0 N + \phi) \end{aligned}$$

which means $\hat{\omega}_0 N = 2\pi m$.
Lets take $\hat{\omega}_0 = 2$, then $n = \pi m \implies \frac{n}{m} = \pi$. So.... not every DT sinusoid is periodic.

Every periodic function (not those irrational edge cases) has a Fundamental period N_0.

$$\begin{gathered} x[n] = A\cos\left(\underbrace{\frac{4\pi}{41}}_{\hat{\omega}_0} n\right) \\ N = \frac{2\pi m}{\frac{4\pi}{41}} = \frac{41m}{2} \end{gathered}$$

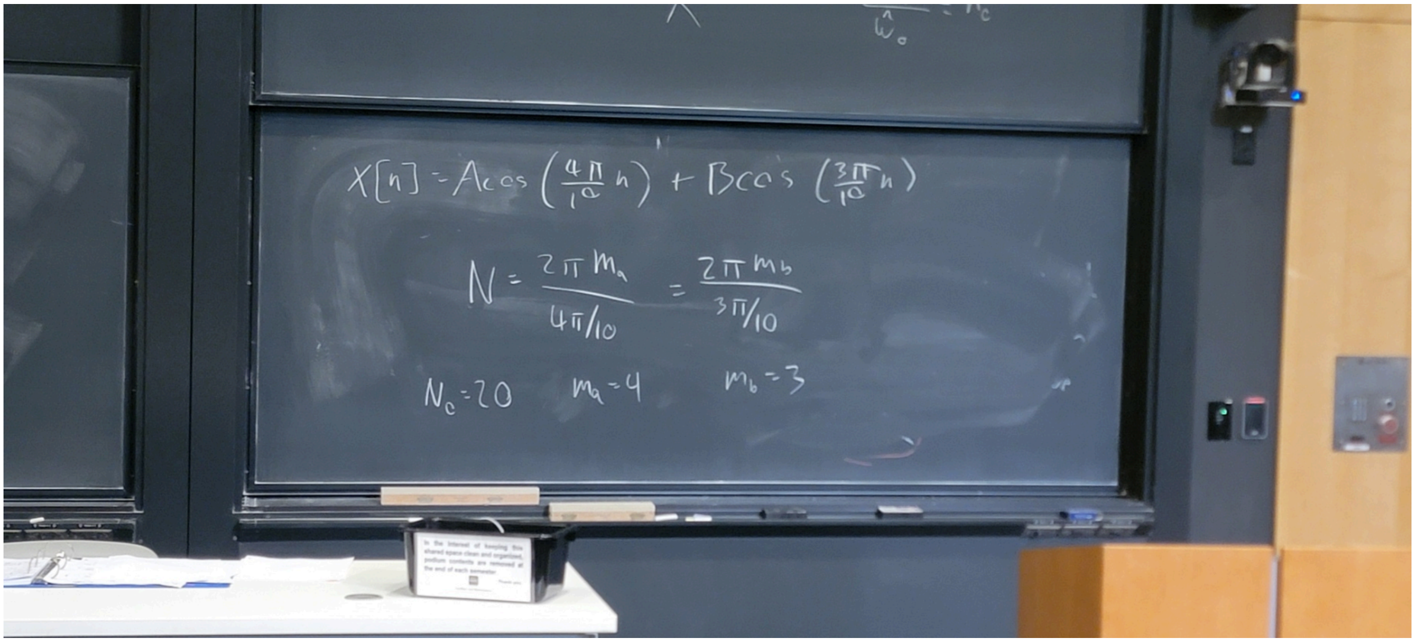
We can pick $m = 2$, $N = 41$.
so $N_0 = 41$, the smallest N for which this is true (where n and m are integers).

What if we have something else:

$$x[n] = A\cos\left(\frac{n}{6}\right)$$
$$N = \frac{2\pi m}{\hat{\omega_0}} = \frac{2\pi m}{6} = \frac{\pi m}{3}$$

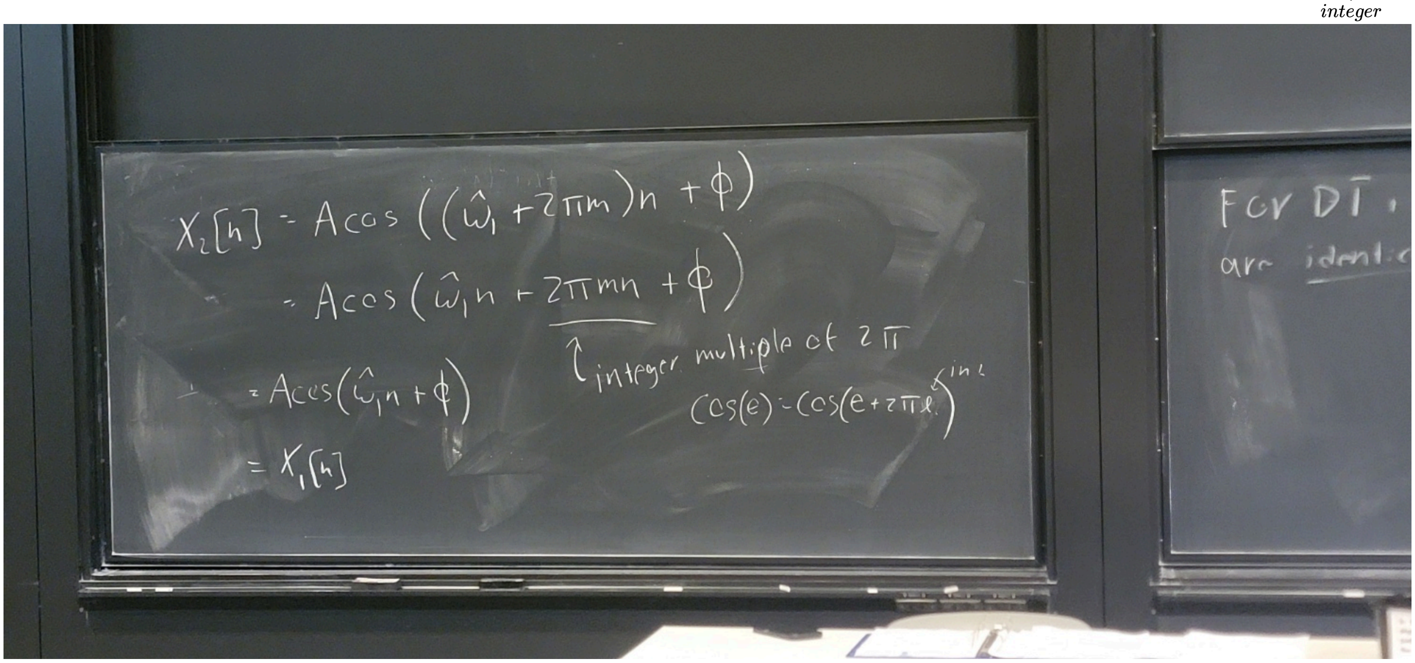
Uh...
$n = \frac{\pi}{3}m$... but n and m must be integers, so this isn't possible. This signal isn't periodic!

One more example



We see that plugging in $N_0 = 20$ gets us π and 6π, which are both multiples of 2π, so that's when these two line up.

For continuous time, different fundamental frequencies are not the same signal. But for discrete time, what if $\hat{\omega_2} = \hat{\omega_1} + \underbrace{2\pi m}_{integer}$



Lets compare

	CT	DT
Dimensions of ω_0	rad/s	rad
Timeshifts	Time shift <-> Phase shift	Timeshift -> Phase shift
Distinct signals	Yes	Identical in some cases when $\hat{\omega_2} = \hat{\omega_1} + 2\pi m$
Periodicity	always	not always

Real exponentials

CT

$$x_t = e^{at} \ -\infty < t < \infty$$

DT

$x[n] = {}^{Bn} = {}^{n}$

We can have exponential growth or decay depending on $< \text{ or } > 1$.

Complex Exponentials

CT: